MHF4U Advanced Functions - Unit 1 Practice Test, Version 6

Answer key and brief explanations

20 questions x 1 mark = 20 marks. Award 1 mark for the single correct option for each question.

Q	Answer	Explanation
1	A	Each input appears with exactly one output; a repeated output is allowed.
2	C	$x+2\geq 0$ and $x-1$ cannot equal zero, so $x\geq -2$ except $x=1$.
3	B	The upward V has minimum -5.
4	D	Exclude -1 and include 4.
5	C	At $x=3$, use the $x\geq 3$ branch: $3^2-4=5$.
6	D	The left value approaches $3(1)+2=5$; require $k(1)-1=5$, giving $k=6$.
7	B	$(-x)^5+2(-x)=-(x^5+2x)$.
8	A	Set $-3(x+2)=-6$, giving $x=0$; new output $2(4)-1=7$.
9	B	The factor 4 inside compresses horizontally by $1/4$; $(x-2)$ moves the graph right 2.
10	C	Swap x and y in $y=4x-7$, then solve: $y=(x+7)/4$.
11	D	$x\geq 2$ selects the plus-root branch: $q^{-1}(x)=2+\sqrt{5-x}$.
12	B	The left x-endpoint -2 is open; the right x-endpoint 3 is filled.
13	A	Centre height $2+5=7$ m; top height $7+5=12$ m.
14	C	After a quarter turn (75 s), the rider is level with the centre: 7 m.
15	B	$(14-2)/(4-1)=12/3=4$.
16	C	$f(1)=9$, $f(3)=21$; $(21-9)/(3-1)=6$.
17	B	$f(1.9)=-7.79$, $f(2.1)=-8.19$; $(-8.19+7.79)/0.2=-2$.
18	C	Secant slope $(0-4)/(3-1)=-2$.
19	A	The tangent at the vertex A is horizontal, so its slope is 0.
20	D	$-5\leq 2x-3\leq 5$ becomes $-1\leq x\leq 4$, or $[-1,4]$.